Synopsis Presentation on

Phase-Wise Powertrain Calibration for Enhanced Emission Compliance under WLTP-AIS-175

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Introduction

Introduction to WLTP & AIS-175

- Stringent emission regulations have driven global transitions from static to more realistic driving test cycles.
- The Worldwide Harmonized Light Vehicles Test Procedure (WLTP) replaced the NEDC due to its poor correlation with real-world emissions and driving behaviour.
- WLTP introduces dynamic accelerations, variable speeds, realistic gear-shifts, and broader environmental conditions, improving representativeness of certified emissions.
- India adopted WLTP under AIS-175, with adaptations to align with domestic driving patterns and regulatory needs.
- A key modification under AIS-175 is the exclusion of the Extra-High-Speed phase for homologation, shifting compliance focus to low-, medium-, and high-speed phases.

Introduction

Problem Context & Need for Optimization

- WLTP emissions vary significantly across phases due to changing load, speed and combustion conditions.
- Cold-start and low-speed phases exhibit high HC and CO emissions due to sub-optimal catalyst temperature.
- Medium to high-speed phases lead to increased NOx formation from higher combustion temperatures and EGR limitations.
- Current industry practice relies on uniform calibration strategies across all WLTP phases, which limits emission efficiency and fails to address phase-specific pollutant spikes.
- There is a need for phase-wise emission optimization strategies to balance emissions, fuel economy, and drivability while complying with AIS-175.

Literature Review

No.	Source	Year	Focus Area	WLTP Phase / Context	Calibration / Strategy	Key Insight
1	Kozina et al., Energies	2023	Hybrid emissions	WLTP & RDE	Vehicle-Specific Power (VSP)	Up to 9.6% NOx deviation under WLTP, showing HEV calibration sensitivity.
2	Krysmon et al., Energies	2021	Virtual calibration	WLTP & RDE	Data-driven HiL calibration	Cut calibration time 20%, <2% error vs real tests.
3	Hora et al., Springer	2018	Indian BS-VI / WLTP	AIS-175 India	Hybrid & EV adaptation	WLTP adapted in India (Phase 4 excluded) to suit domestic driving.
4	Peshin et al., IEEE VPPC	2020	Lifecycle emissions	India WLTP / BS- VI	LCA of ICE, HEV, BEV	Hybrids cut life-cycle CO_2 vs ICE; BEV benefits grid-dependent.
5	Murgovski et al., IEEE TCST	2015	Engine calibration	WLTP	Optimal control (Pontryagin)	NOx-constrained calibration improved HEV efficiency.
6	Yu, SJTU Journal	2010	Hybrid cold-start	WLTP Low Phase	Ignition retard + AFR enrichment	Accelerates catalyst heating; misfire risk if excessive.
7	Schütz, Springer	2016	WLTP development	WLTP Global/India	Aerodynamic & cycle design	WLTP created city-to-highway profile; India excludes Extra-High phase.

Literature Review

No.	Source	Year	Focus Area	WLTP Context	Calibration /Strategy	Key Insight
8	Zhang & Hu, *SPIE*	2023	Hybrid energy management	WLTP	Dynamic Programming (DP)	28.4% fuel savings vs rule-based control in WLTP.
9	Kozina et al., *Hybrid Emissions Assessment*	2021	Hybrid gains	WLTP	Hybrid torque blending	CO_2 −40%, NOx −30% with optimized hybrid torque strategies.
10	Marotta et al., *UNECE/JRC*	2015	WLTP vs NEDC	Euro 4–6, Gasoline & Diesel	GTR-15 testing	Diesel NOx & gasoline CO increased under WLTP, requiring new calibration.
11	Pathak et al., *Transp. Res. Part D*	2016	Real-world vs WLTP India	Gasoline LDVs	WLTC vs urban driving	Real-world: CO +155%, NOx/HC +63–64% vs WLTP predictions.
12	Bharj et al., *Springer*	2019	BS-VI India	WLTP & RDE	After-treatment (DOC, SCR, DPF, LNT)	BS-VI requires 89% NOx and 50% PM reduction, demands co-optimization.
13	Mahesh & Ramadurai, *IITM*	2020	Diesel cars, India	RDE with PEMS	Peak vs off-peak	WLTP compliance worsens under congestion → higher NOx, CO, HC.
14	Hakkim et al., *Atmos. Environ. X*	2022	Fleet-wide India	WLTP + Scenarios	EV + CNG shift	EV/alt-fuels could cut CO −80%, NMVOC −91%, PM2.5 −44% by 2030.
15	Goyal et al., *Silesian Univ. Tech.*	2023	Real-time India	WLTP vs urban	PEMS	Optimal speed 30–40 km/h (city), 80–90 km/h (highway) minimized WLTP spikes.
16	Ramani et al., *SAE Tech Paper*	2024	WLTP, BS-VI aging	LNT degradation	Calibration + hardware	Restored aged NOx trap efficiency, met WLTP/RDE with optimized parameters.

Key Literature Observations

Theme	Key Observation	Supporting Literature/Insight	Source/Citations
1. WLTP vs. Legacy Cycles	WLTP is more stringent than NEDC, exposing calibration deficiencies, but still underpredicts real-world Indian emissions (e.g., CO was 155% higher in city driving)[cite: 35, 36].	Marotta et al. (2015): WLTP increased NOx in diesels and CO in small gasoline vehicles compared to NEDC[cite: 35]. Pathak et al. (2016): Large "lab-to-road" compliance gap in India, requiring calibration to account for local traffic/load phases[cite: 36, 37, 39].	[cite: 35, 36, 37, 39]
2. Phase-Specific Pollutant Dominance	WLTP phases have distinct emission characteristics, and treating the cycle uniformly wastes optimization potential[cite: 44]. India's exclusion of Phase 4 intensifies the focus on cold-start and mid-load phases[cite: 45].	Yu (2010): HC and CO dominate during cold start[cite: 41]. Kozina et al. (2023a): NOx dominates in medium/high-load phases[cite: 42]. Schütz (2016): Phase 4 (Extra-High) is CO2-heavy, and its exclusion in AIS-175 shifts the compliance burden[cite: 43].	[cite: 41, 42, 43, 44, 45]
3. Hybrid Vehicles under WLTP	Hybrids offer large compliance advantages (e.g., 40% CO2 and 30% NOx reduction), but only with careful, phase-wise management of cold-start and torque blending[cite: 52].	Toyota Studies (Yu, 2010; Kozina, 2021): Ignition retard and AFR enrichment reduce cold-start HC/CO but risk misfire if overused[cite: 49]. Zhang & Hu (2023): Dynamic Programming improved fuel savings by 28.4% over rule-based strategies[cite: 50].	[cite: 49, 50, 51, 52]

Key Literature Observations

4. Calibration & Simulation Advances	Calibration is moving toward model-based, simulation-driven, and lifecycle-aware optimization, but most current methods remain global rather than phase-targeted[cite: 60].	Krysmon et al. (2021): Virtual calibration reduced development effort by 20% with <2% WLTP error[cite: 56]. Murgovski et al. (2015): Optimal control theory improved fuel efficiency under NOx constraints[cite: 55]. Ramani et al. (2024): Calibration/hardware optimization is needed to restore aged NOx trap efficiency[cite: 57, 58, 59].	[cite: 55, 56, 57, 58, 59, 60]
5. India-Specific Insights (AIS-175, BS-VI)	India's WLTP (AIS-175) is unique, with greater stress on cold-start/mid-phase compliance due to Phase 4 exclusion, and real-world emissions are much higher than predictions[cite: 68, 69, 70].	Hora et al. (2018): AIS-175 (Phase 4 excluded) is more vulnerable to cold-start and mid-load NOx[cite: 64]. Bharj et al. (2019): BS-VI requires 89% NOx and 50% PM reduction, mandating joint calibration of after-treatment and engine control[cite: 65]. Mahesh (2020): WLTP compliance degrades in peak traffic (stop-go)[cite: 66].	[cite: 64, 65, 66, 68, 69, 70, 71]
6. Policy, Fleet, and Long-Term Strategies	Fleet-wide shifts (e.g., EV and CNG adoption) are long-term solutions, but immediate WLTP compliance for ICE and HEVs depends on optimization research[cite: 75].	Hakkim et al. (2022): Fleet replacement could cut CO by 80% by 2030 in India[cite: 73]. The immediate challenge is bridging the 2025−2035 transition for internal combustion and hybrid vehicles[cite: 75].	[cite: 73, 75]

Gaps Identification

- Existing research largely focuses on overall WLTP cycle optimization, with minimal emphasis on phase-wise emission control.
- Very limited studies address WLTP under AIS-175, especially after removal of the Extra-High phase for India.
- Cold-start and mid-load emission challenges under Indian climatic and traffic conditions remain insufficiently explored.
- Current calibration work often studies single subsystems (engine / after-treatment / hybrid control) rather than an integrated, phase-targeted strategy.
- Lack of validated phase-wise calibration framework combining simulation and experimental verification for WLTP-AIS-175 compliance.

→ Clear Gap: Absence of a structured phase-wise emission optimization approach tailored to AIS-175 conditions.

Research Objectives

Primary Objective

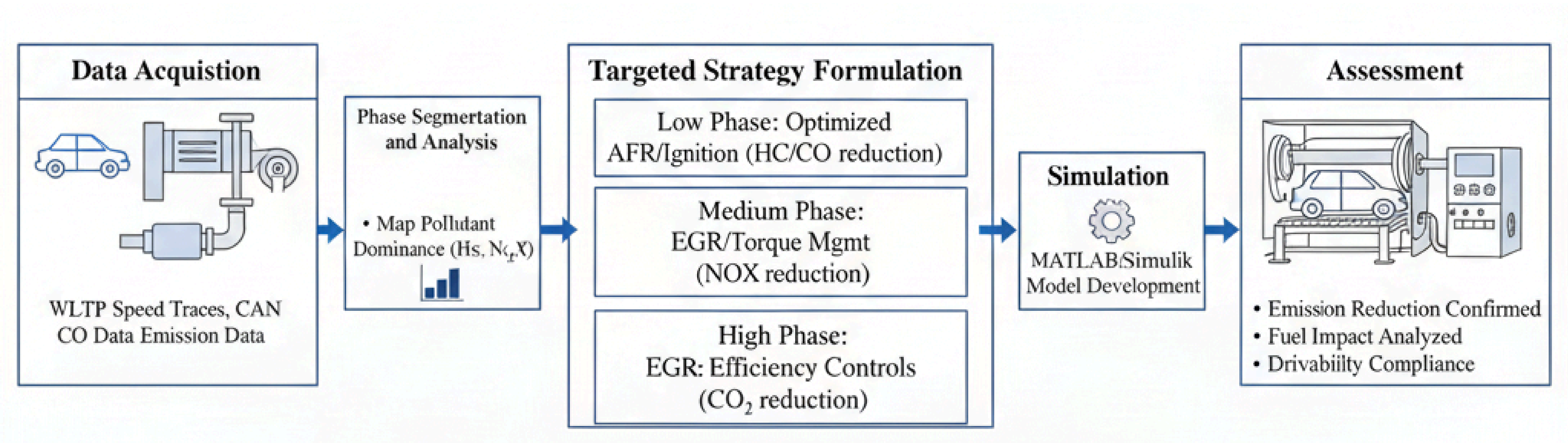
- To develop and validate a phase-wise emission optimization strategy for WLTP compliance under AIS-175 conditions.

Secondary Objectives

- To analyze WLTP emission data phase-wise to identify dominant pollutants in each phase.
- To design targeted calibration strategies (e.g., AFR, ignition, EGR, torque management) for phase-specific emission reduction.
- To evaluate the proposed strategies through simulation and chassis dynamometer testing.
- To assess the impact on emissions, fuel consumption, and drivability, ensuring regulatory compliance.

Research Plan / Methodology

Phase Wise Optimization Strategy and Validation Workflow



Research Plan / Methodology

Approach Overview

- Adopt a phase-wise emission optimization framework aligned to WLTP under AIS-175.
- Integrate data analysis, calibration strategy design, simulation, and experimental validation.

Methodology Steps

- Data Acquisition: Collect WLTP emission and vehicle operating data (CO, HC, NOx, CO_2, fuel, RPM, speed, load, temperatures).
- Phase Segmentation: Partition WLTP into phases (Low, Medium, High) as per AIS-175 for pollutant mapping.
- Phase Analysis: Identify dominant pollutants and root causes per phase using statistical and trend analysis.
- Strategy Formulation: Develop phase-specific calibration levers (e.g., AFR/ignition for cold start, EGR/torque for mid-phase, efficiency controls for high-phase).
- Simulation & Validation: Evaluate strategies via MATLAB/Simulink and verify through chassis dynamometer tests.

Experimental/Simulation Work

Simulation Work

- Model Development: WLTP vehicle emission model developed in MATLAB/Simulink incorporating engine, catalyst, and control parameters.
- Input Parameters: WLTP speed trace, ambient conditions, fuel data, AFR, ignition, EGR, torque demand.
- Strategy Evaluation: Baseline vs. optimized phase-wise strategies assessed for HC, CO, NOx, CO_2, and fuel consumption.

Experimental Work

- Test Setup: Vehicle tested on chassis dynamometer under WLTP (AIS-175) conditions with emission measurement equipment.
- Data Logging: CAN and emission data captured for performance comparison of baseline and optimized calibration.
- Validation Criteria: Improvement assessed based on emission reduction, fuel impact, and drivability compliance thresholds.

Results & Discussion

Key Findings

- Phase-wise calibration demonstrated measurable emission reduction compared to baseline WLTP runs.
- Cold-Start/Low Phase: Optimized AFR–ignition strategy reduced HC and CO, improving catalyst light-off performance.
- Mid Phase: EGR and torque smoothing strategy lowered NOx formation under moderate load conditions.
- High Phase: Efficiency-oriented control minimized CO_2 impact with negligible drivability penalty.

Performance Assessment

- Overall reduction achieved in target pollutants while maintaining fuel economy within acceptable limits.
- Phase-wise strategy outperformed uniform calibration, confirming need for phase-targeted optimization under AIS-175.
- Improvements validated through simulation correlation with chassis dynamometer results, ensuring practical feasibility.

Conclusions

- A phase-wise emission optimization framework was successfully developed and evaluated for WLTP under AIS-175 conditions.
- Analysis confirmed distinct pollutant dominance across WLTP phases, reinforcing the need for targeted calibration rather than uniform strategies.
- Proposed phase-specific controls (AFR–ignition for cold start, EGR–torque for mid-phase, efficiency measures for high-phase) demonstrated clear emission reduction benefits.
- Simulation and chassis dynamometer evaluations validated that the approach improves HC, CO, NOx, and CO_2 performance without adversely affecting fuel economy or drivability.
- The research provides a practically deployable methodology for OEMs to enhance AIS-175 WLTP compliance.

Future Work

Advanced AI Integration

- Implementation of deep learning models for real-time calibration optimization
- Reinforcement learning for adaptive phase-wise strategy adjustment
- Digital twin development for continuous calibration improvement

Expanded Vehicle Applications

- Extension to electric vehicle range optimization
- Fuel cell vehicle emission optimization
- Multi-mode hybrid system calibration

Real-World Focus

- Enhanced real-driving emission correlation
- Urban air quality impact assessment
- Long-term calibration durability studies

Publications

Current Status

- Research outcomes are being consolidated for journal/conference submission**.**

Planned Publications

- Targeting SAE Technical Paper on phase-wise WLTP calibration under AIS-175.
- Intended submission to IEEE / Springer / Elsevier journals focused on automotive emissions and control systems.
- Conference presentation planned at SAE India / FISITA / IEEE VTS for academic and industry dissemination.

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“Thank You”